

Tangent half-angle substitution

The **tangent half-angle substitution** is a change of variables used for evaluating integrals, which converts a rational function of trigonometric functions of x into an ordinary rational function of t by setting $t = \tan \frac{x}{2}$. This is the one-dimensional stereographic projection of the unit circle parametrized by angle measure onto the real line. The general^[1] transformation formula is:

$$\int f(\sin x, \cos x) dx = \int f\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right) \frac{2 dt}{1+t^2}.$$

The tangent of half an angle is important in spherical trigonometry and was sometimes known in the 17th century as the half tangent or semi-tangent.^[2] Leonhard Euler used it to evaluate the integral $\int dx/(a + b \cos x)$ in his 1768 integral calculus textbook,^[3] and Adrien-Marie Legendre described the general method in 1817.^[4]

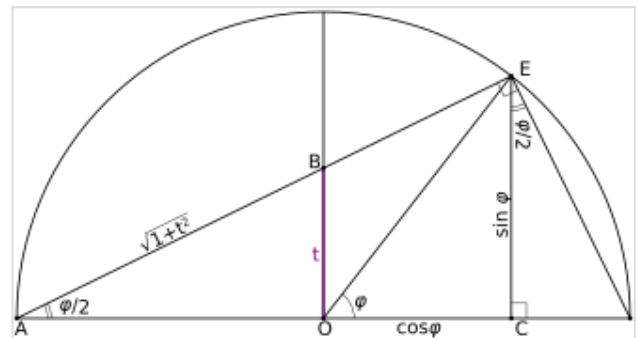
The substitution is described in most integral calculus textbooks since the late 19th century, usually without any special name.^[5] It is known in Russia as the **universal trigonometric substitution**,^[6] and also known by variant names such as *half-tangent substitution* or *half-angle substitution*. It is sometimes misattributed as the **Weierstrass substitution**.^[7] Michael Spivak called it the "world's sneakiest substitution".^[8]

The substitution

Introducing a new variable $t = \tan \frac{x}{2}$, sines and cosines can be expressed as rational functions of t , and dx can be expressed as the product of dt and a rational function of t , as follows:

$$\sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}, \quad \text{and} \quad dx = \frac{2 dt}{1+t^2}.$$

Similar expressions can be written for $\tan x$, $\cot x$, $\sec x$, and $\csc x$.



The tangent half-angle substitution relates an angle to the slope of a line.

Derivation

Using the double-angle formulas $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$ and $\cos x = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}$ and introducing denominators equal to one by the Pythagorean identity $1 = \cos^2 \frac{x}{2} + \sin^2 \frac{x}{2}$ results in

$$\sin x = \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2}} = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} = \frac{2t}{1 + t^2},$$

$$\cos x = \frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2}} = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} = \frac{1 - t^2}{1 + t^2}.$$

Finally, since $t = \tan \frac{x}{2}$, differentiation rules imply

$$dt = \frac{1}{2} (1 + \tan^2 \frac{x}{2}) dx = \frac{1 + t^2}{2} dx,$$

and thus

$$dx = \frac{2}{1 + t^2} dt.$$

Examples

Antiderivative of cosecant

$$\begin{aligned} \int \csc x \, dx &= \int \frac{dx}{\sin x} \\ &= \int \left(\frac{1 + t^2}{2t} \right) \left(\frac{2}{1 + t^2} \right) dt \quad t = \tan \frac{x}{2} \\ &= \int \frac{dt}{t} \\ &= \ln |t| + C \\ &= \ln \left| \tan \frac{x}{2} \right| + C. \end{aligned}$$

We can confirm the above result using a standard method of evaluating the cosecant integral by multiplying the numerator and denominator by $\csc x - \cot x$ and performing the substitution $u = \csc x - \cot x$, $du = (-\csc x \cot x + \csc^2 x) dx$.

$$\begin{aligned}
\int \csc x \, dx &= \int \frac{\csc x (\csc x - \cot x)}{\csc x - \cot x} \, dx \\
&= \int \frac{(\csc^2 x - \csc x \cot x) \, dx}{\csc x - \cot x} & u &= \csc x - \cot x \\
&= \int \frac{du}{u} \\
&= \ln |u| + C \\
&= \ln |\csc x - \cot x| + C.
\end{aligned}$$

These two answers are the same because $\csc x - \cot x = \tan \frac{x}{2}$:

$$\begin{aligned}
\csc x - \cot x &= \frac{1}{\sin x} - \frac{\cos x}{\sin x} \\
&= \frac{1+t^2}{2t} - \frac{1-t^2}{1+t^2} \frac{1+t^2}{2t} & t &= \tan \frac{x}{2} \\
&= \frac{2t^2}{2t} = t \\
&= \tan \frac{x}{2}
\end{aligned}$$

The secant integral may be evaluated in a similar manner.

A definite integral

We wish to evaluate the integral:

$$\int_0^{2\pi} \frac{dx}{2 + \cos x}$$

A naïve approach splits the interval and applies the substitution $t = \tan \frac{x}{2}$. However, this substitution has a singularity at $x = \pi$, which corresponds to a vertical asymptote. Therefore, the integral must be split at that point and handled carefully:

$$\begin{aligned}
\int_0^{2\pi} \frac{dx}{2 + \cos x} &= \int_0^{\pi} \frac{dx}{2 + \cos x} + \int_{\pi}^{2\pi} \frac{dx}{2 + \cos x} \\
&= \int_0^{\infty} \frac{2 dt}{3 + t^2} + \int_{-\infty}^0 \frac{2 dt}{3 + t^2} \quad \text{where } t = \tan \frac{x}{2} \\
&= \int_{-\infty}^{\infty} \frac{2 dt}{3 + t^2} \\
&= \frac{2}{\sqrt{3}} \int_{-\infty}^{\infty} \frac{du}{1 + u^2} \quad \text{where } t = u\sqrt{3} \\
&= \frac{2\pi}{\sqrt{3}}.
\end{aligned}$$

Note: The substitution

$t = \tan \frac{x}{2}$ maps $x \in (0, \pi)$ to $t \in (0, \infty)$ and $x \in (\pi, 2\pi)$ to $t \in (-\infty, 0)$. The point $x = \pi$ corresponds to a vertical asymptote in t , so the integral is evaluated as a limit around this point.

Alternatively, we can compute the indefinite integral first:

$$\int \frac{dx}{2 + \cos x} = \frac{2}{\sqrt{3}} \arctan\left(\frac{\tan \frac{x}{2}}{\sqrt{3}}\right) + C \quad \text{Using symmetry:}$$

$$\begin{aligned}
\int_0^{2\pi} \frac{dx}{2 + \cos x} &= 2 \int_0^{\pi} \frac{dx}{2 + \cos x} \\
&= 2 \lim_{b \rightarrow \pi^-} \left[\frac{2}{\sqrt{3}} \arctan\left(\frac{\tan \frac{x}{2}}{\sqrt{3}}\right) \right]_0^b \quad \text{Thus, the value of the definite integral is:} \\
&= \frac{4}{\sqrt{3}} \left(\frac{\pi}{2} - 0 \right) = \frac{2\pi}{\sqrt{3}}.
\end{aligned}$$

$$\int_0^{2\pi} \frac{dx}{2 + \cos x} = \frac{2\pi}{\sqrt{3}}$$

Third example: both sine and cosine

$$\begin{aligned}
\int \frac{dx}{a \cos x + b \sin x + c} &= \int \frac{2 dt}{a(1 - t^2) + 2bt + c(t^2 + 1)} \\
&= \int \frac{2 dt}{(c - a)t^2 + 2bt + a + c} \\
&= \frac{2}{\sqrt{c^2 - (a^2 + b^2)}} \arctan\left(\frac{(c - a) \tan \frac{x}{2} + b}{\sqrt{c^2 - (a^2 + b^2)}}\right) + C
\end{aligned}$$

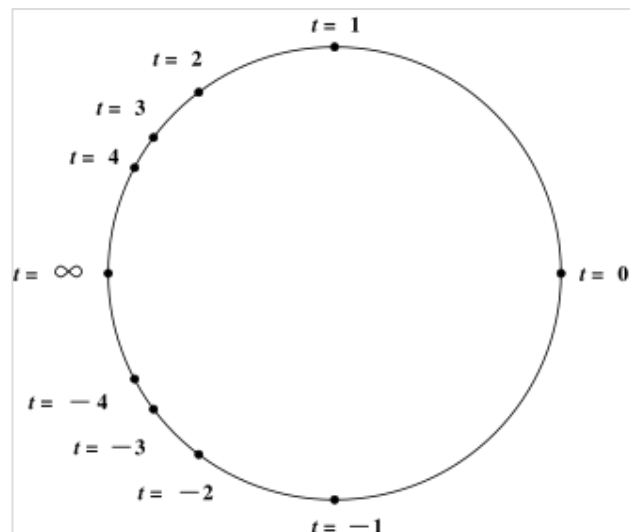
if $c^2 - (a^2 + b^2) > 0$.

Geometry

As x varies, the point $(\cos x, \sin x)$ winds repeatedly around the unit circle centered at $(0, 0)$. The point

$$\left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right)$$

goes only once around the circle as t goes from $-\infty$ to $+\infty$, and never reaches the point $(-1, 0)$, which is approached as a limit as t approaches $\pm\infty$. As t goes from $-\infty$ to -1 , the point determined by t goes through the part of the circle in the third quadrant, from $(-1, 0)$ to $(0, -1)$. As t goes from -1 to 0 , the point follows the part of the circle in the fourth quadrant from $(0, -1)$ to $(1, 0)$. As t goes from 0 to 1 , the point follows the part of the circle in the first quadrant from $(1, 0)$ to $(0, 1)$. Finally, as t goes from 1 to $+\infty$, the point follows the part of the circle in the second quadrant from $(0, 1)$ to $(-1, 0)$.



The tangent half-angle substitution parametrizes the unit circle centered at $(0, 0)$. Instead of $+\infty$ and $-\infty$, we have only one ∞ , at both ends of the real line. That is often appropriate when dealing with rational functions and with trigonometric functions. (This is the one-point compactification of the line.)

Here is another geometric point of view. Draw the unit circle, and let P be the point $(-1, 0)$. A line through P (except the vertical line) is determined by its slope. Furthermore, each of the lines (except the vertical line) intersects the unit circle in exactly two points, one of which is P . This determines a function from points on the unit circle to slopes. The trigonometric functions determine a function from angles to points on the unit circle, and by combining these two functions we have a function from angles to slopes.

Hyperbolic functions

As with other properties shared between the trigonometric functions and the hyperbolic functions, it is possible to use hyperbolic identities to construct a similar form of the substitution, $t = \tanh \frac{x}{2}$:

$$\sinh x = \frac{2t}{1 - t^2}, \quad \cosh x = \frac{1 + t^2}{1 - t^2}, \quad \text{and} \quad dx = \frac{2}{1 - t^2} dt.$$

Similar expressions can be written for $\tanh x$, $\coth x$, $\operatorname{sech} x$, and $\operatorname{csch} x$. Geometrically, this change of variables is a one-dimensional stereographic projection of the hyperbolic line onto the real interval, analogous to the Poincaré disk model of the hyperbolic plane.

Alternatives

There are other approaches to integrating trigonometric functions. For example, it can be helpful to rewrite trigonometric functions in terms of e^{ix} and e^{-ix} using Euler's formula.

See also



- Rational curve
- Stereographic projection
- Tangent half-angle formula
- Trigonometric substitution
- Euler substitution

Further reading

- Courant, Richard (1937) [1934]. "1.4.6. Integration of Some Other Classes of Functions §1–3" (<https://archive.org/details/differentialinte0001cour/page/234>). *Differential and Integral Calculus*. Vol. 1. Blackie & Son. pp. 234–237.
- Edwards, Joseph (1921). "§1.6.193" (<https://archive.org/details/treatiseonintegr01edwauoft/page/187>). *A Treatise on the Integral Calculus*. Vol. 1. Macmillan. pp. 187–188.
- Hardy, Godfrey Harold (1905). "VI. Transcendental functions" (<https://archive.org/details/integrationoffun00hardrich/page/42>). *The integration of functions of a single variable*. Cambridge. pp. 42–51. Second edition 1916, pp. 52–62 (<https://archive.org/details/integrationoffun00hard/page/52/>)
- Hermite, Charles (1873). "Intégration des fonctions transcendentes" (<https://archive.org/details/coursdanalysedel01hermuoft/page/320>) [Integration of transcendental functions]. *Cours d'analyse de l'école polytechnique* (in French). Vol. 1. Gauthier-Villars. pp. 320–380.
- Stewart, Seán M. (2017). "14. Tangent Half-Angle Substitution". *How to Integrate It*. Cambridge. pp. 178–187. doi:10.1017/9781108291507.015 (<https://doi.org/10.1017%2F9781108291507.015>). ISBN 978-1-108-41881-2.

Notes and references

1. Other trigonometric functions can be written in terms of sine and cosine.
2. Gunter, Edmund (1673) [1624]. *The Works of Edmund Gunter* (<https://archive.org/details/worksofedmundgun00gunt/>). Francis Eglesfield. p. 73 (<https://archive.org/details/worksofedmundgun00gunt/page/n104/>)

3. Euler, Leonhard (1768). "§1.1.5.261 Problema 29" (<http://eulerarchive.maa.org/docs/originals/E342sec1ch5.pdf>) (PDF). *Institutiones calculi integralis* [Foundations of Integral Calculus] (in Latin). Vol. I. Impensis Academiae Imperialis Scientiarum. pp. 148–150. E342 (<https://scholarlycommons.pacific.edu/euler-works/342/>), Translation by Ian Bruce (<http://www.17centurymaths.com/contents/euler/intcalvol1/part1ch5.pdf>). Also see Lobatto, Rehuel (1832). "19. Note sur l'intégration de la fonction $\partial z / (a + b \cos z)$ " (<https://archive.org/details/journalfurdierei9101unse/page/n268>). *Crelle's Journal* (in French). **9**: 259–260.
4. Legendre, Adrien-Marie (1817). *Exercices de calcul intégral* (<https://archive.org/details/exercicescalculi02legerich>) [*Exercises in integral calculus*] (in French). Vol. 2. Courcier. p. 245–246 (<https://archive.org/details/exercicescalculi02legerich/page/n267/>).
5. For example, in chronological order,
 - Hermite, Charles (1873). *Cours d'analyse de l'école polytechnique* (<https://archive.org/details/coursdanalysedel01hermuoft/page/320/>) (in French). Vol. 1. Gauthier-Villars. p. 320.
 - Johnson, William Woolsey (1883). *An Elementary Treatise on the Integral Calculus* (<http://archive.org/details/anelementarytre00johngoog/page/n66>). MacMillan. p. 52.
 - Picard, Émile (1901) [1891]. *Traité d'analyse* (<https://archive.org/details/traitedanalyse03picagooog/page/77>) (in French). Vol. 1 (2e ed.). Gauthier-Villars. p. 77.
 - Goursat, Édouard (1904) [1902]. *A Course in Mathematical Analysis* (<https://archive.org/details/courseinmathemat01gouruoft/page/236>). Vol. 1. Translated by Hedrick, Earle Raymond. Ginn. pp. 236ff.
 - Wilson, Edwin Bidwell (1911). *Advanced Calculus* (<https://archive.org/details/advancedcalculus00wils/page/21/>). Ginn. p. 21.
 - Edwards, Joseph (1921). *A Treatise on the Integral Calculus* (<https://archive.org/details/treatiseonintegr01edwauoft/page/188>). Vol. 1. MacMillan. pp. 187–188.
 - Courant, Richard (1937) [1930]. *Differential and Integral Calculus* (<https://archive.org/details/ost-math-courant-differentialintegralcalculusvoli/page/n249/>). Vol. 1. Translated by McShane, E. J. (2nd ed.). Blackie & Son. pp. 234–238.
 - Peterson, Thurman S. (1950). *Elements of Calculus* (<https://archive.org/details/ofcalcul00pete/page/201/>). Harper & Brothers. pp. 201–202.
 - Apostol, Tom M. (1967) [1961]. *Calculus* (<https://archive.org/details/calculus0000apos/page/264/>). Vol. 1 (2nd ed.). Xerox. pp. 264–265.
 - Larson, Ron; Hostetler, Robert P.; Edwards, Bruce H. (1998) [1978]. *Calculus of a Single Variable* (<https://archive.org/details/calculusofsingle00lars/page/520>) (6th ed.). Houghton Mifflin. p. 520. ISBN 9780395885789.
 - Rogawski, Jon (2011) [2008]. *Calculus: Early Transcendentals* (<https://books.google.com/books?id=rn4paEb8izYC&pg=PA435>) (2nd ed.). Macmillan. p. 435. ISBN 9781429231848.
6. Piskunov, Nikolai (1969). *Differential and Integral Calculus* (<https://archive.org/details/n.-piskunov-differential-and-integral-calculus-mir-1969/page/379/>). Mir. p. 379.
Zaitsev, V. V.; Ryzhkov, V. V.; Skanavi, M. I. (1978). *Elementary Mathematics: A Review Course* (<https://catalog.hathitrust.org/Record/000248796>). *Elementarnaiâ matematika*. English. Mir. p. 388.

7. In 1966 William Eberlein attributed this substitution to Karl Weierstrass (1815–1897):

Eberlein, William Frederick (1966). "The Circular Function(s)". *Mathematics Magazine*. **39** (4): 197–201. doi:10.1080/0025570X.1966.11975715 (<https://doi.org/10.1080%2F0025570X.1966.11975715>). JSTOR 2688079 (<https://www.jstor.org/stable/2688079>). "(Equations (3) [$x = \cos \theta$], (4) [$y = \sin \theta$], (5) [$t = \tan \frac{\theta}{2}$] are, of course, the familiar half-angle substitutions introduced by Weierstrass to integrate rational functions of sine, cosine.)"

Two decades later, James Stewart mentioned Weierstrass when discussing the substitution in his popular calculus textbook, first published in 1987:

Stewart, James (1987). "§7.5 Rationalizing substitutions" (https://archive.org/details/calculus00stew_3/page/431). *Calculus*. Brooks/Cole. p. 431. ISBN 9780534066901. "The German mathematician Karl Weierstrass (1815–1897) noticed that the substitution $t = \tan(x/2)$ will convert any rational function of $\sin x$ and $\cos x$ into an ordinary rational function."

Later authors, citing Stewart, have sometimes referred to this as the *Weierstrass substitution*, for instance:

- Jeffrey, David J.; Rich, Albert D. (1994). "The evaluation of trigonometric integrals avoiding spurious discontinuities" (<https://doi.org/10.1145%2F174603.174409>). *Transactions on Mathematical Software*. **20** (1): 124–135. doi:10.1145/174603.174409 (<https://doi.org/10.1145%2F174603.174409>). S2CID 13891212 (<https://api.semanticscholar.org/CorpusID:13891212>).
- Merlet, Jean-Pierre (2004). "A Note on the History of Trigonometric Functions" (https://ia800201.us.archive.org/15/items/springer_10.1007-1-4020-2204-2/10.1007-1-4020-2204-2.pdf#page=205) (PDF). In Ceccarelli, Marco (ed.). *International Symposium on History of Machines and Mechanisms*. Kluwer. pp. 195–200. doi:10.1007/1-4020-2204-2_16 (https://doi.org/10.1007%2F1-4020-2204-2_16). ISBN 978-1-4020-2203-6.
- Weisstein, Eric W. (2011). "Weierstrass Substitution" (<https://mathworld.wolfram.com/WeierstrassSubstitution.html>). *MathWorld*. Retrieved 2020-04-01.

Neither Eberlein nor Stewart provided any evidence for the attribution to Weierstrass. A related substitution appears in Weierstrass's *Mathematical Works*, from an 1875 lecture wherein Weierstrass credits Carl Gauss (1818) with the idea of solving an integral of the form $\int d\psi H(\sin \psi, \cos \psi) / \sqrt{G(\sin \psi, \cos \psi)}$ by the substitution $t = -\cot(\psi/2)$.

Weierstrass, Karl (1915) [1875]. "8. Bestimmung des Integrals ..." (<https://archive.org/detail/s/mathematischewer06weieuoft/page/89>). *Mathematische Werke von Karl Weierstrass* (in German). Vol. 6. Mayer & Müller. pp. 89–99.

8. Spivak, Michael (1967). "Ch. 9, problems 9–10" (<https://archive.org/details/CalculusSpivak/page/n337/>). *Calculus*. Benjamin. pp. 325–326.

External links

- Weierstrass substitution formulas (<https://planetmath.org/WeierstrassSubstitutionFormulas>) at PlanetMath

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